1. (a)

The signal $f(x,y) - g(x,y)$ is obtained by the Inverse DFT of the signal $F(u,v) - G(u,v)$. Therefore, according to Parseval's theorem the energy of the signal $f(x,y) - g(x,y)$ is equal to the energy of the signal $F(u,v) - G(u,v)$. The signal $F(u,v) - G(u,v)$ consists of the DFT samples of $F(u,v)$ which were excluded in forming $G(u,v)$. Since, $G(u,v)$ captures 0.95 of the energy of $F(u,v)$, the signal $F(u,v) - G(u,v)$ will capture 0.05 of the energy of $F(u,v)$. Therefore,

$$\sum_{x=0}^{M-1}\sum_{y=0}^{N-1}(f(x,y) - g(x,y))^2 = 0.05\sum_{x=0}^{M-1}\sum_{y=0}^{N-1}f(x,y)^2$$

[5]

(b)

$$f(x,y) = \sum_{x=0}^{M-1}\sum_{y=0}^{N-1}G(u,v)e^{j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}$$

$$g(x,y) = \sum_{x=0}^{M-1}\sum_{y=0}^{N-1}F(u,v)e^{j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)} - F(u_0,v_0)e^{j2\pi\left(\frac{u_0x}{M}+\frac{v_0y}{N}\right)}$$

$$- F(M-u_0,N-v_0)e^{j2\pi\left(\frac{(M-u_0)x}{M}+\frac{(N-v_0)y}{N}\right)}$$

$$= f(x,y) - F(u_0,v_0)e^{j2\pi\left(\frac{u_0x}{M}+\frac{v_0y}{N}\right)} - F(M-u_0,N-v_0)e^{j2\pi\left(\frac{(M-u_0)x}{M}+\frac{(N-v_0)y}{N}\right)}$$

[5]

(c)

The eigenvalues of the covariance matrix $\underline{C_f} = \begin{bmatrix} a & b & 0 \\ b & a & 0 \\ 0 & 0 & c \end{bmatrix}$ are found by the following relationship:

$$\det\begin{bmatrix} a-\lambda & b & 0 \\ b & a-\lambda & 0 \\ 0 & 0 & c-\lambda \end{bmatrix} = (c-\lambda)[(a-\lambda)^2 - b^2]$$

$$= (c-\lambda)[(a-\lambda)-b](c-\lambda)[(a-\lambda)+b]$$

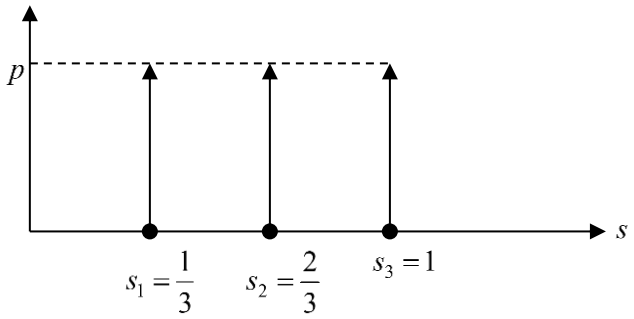
$$= 0 \Rightarrow \lambda_1 = c, \lambda_2 = a-b, \lambda = a+b$$

(i) If $c < a-b$ then because $c \geq 0$ since it represents variance of an image the eigenvalues will be sorted according to magnitude as $a+b \geq a-b > c$ and therefore by using only one principal component the error of reconstruction will be $a-b+c$. **[5]**

(ii) If $c > a+b$ the eigenvalues will be sorted according to magnitude as $c > a+b \geq a-b$ and therefore by using only two principal components the error of reconstruction will be $a-b$. **[5]**

2. (a)

(i) We can use histogram equalization. By doing so, the three intensities are mapped to the following: $s_1 = T(r_1) = \frac{1}{3}$, $s_2 = T(r_2) = \frac{2}{3}$, $s_3 = T(r_3) = 1$ **[5]**



(ii) For the original image we have:

Mean: $m_1 = \frac{1}{3}(r_1 + r_2 + r_3)$

Variance: $\sigma_1^2 = \frac{1}{3}(r_1^2 + r_2^2 + r_3^2) - \frac{1}{9}(r_1 + r_2 + r_3)^2$

For the equalised image we have:

Mean: $m_2 = \frac{1}{3}(s_1 + s_2 + s_3) = \frac{1}{3}(\frac{1}{3} + \frac{2}{3} + 1) = \frac{2}{3}$

Variance: $\sigma_2^2 = \frac{1}{3}(s_1^2 + s_2^2 + s_3^2) - \frac{4}{9} = \frac{1}{3}(\frac{1}{9} + \frac{4}{9} + 1) - \frac{4}{9} = \frac{2}{27}$ **[5]**

(b)

Since dark areas are represented by small intensities, we can calculate the local mean around each pixel in order to decide whether this belongs to a dark or to a bright area. We denote the mean around a pixel with co-ordinates (x, y) as $m_{(x,y),W}$. The subscript W denotes the neighborhood used in order to calculate this mean. If this is a rectangular window of size $(2W + 1)(2W + 1)$ symmetrically placed around the pixel of interest, then

$$m_{(x,y),W} = \frac{1}{(2W+1)^2} \sum_{i=-W}^{i=+W} \sum_{j=-W}^{j=+W} f(x + i, y + j)$$

We can calculate the local variance around each pixel in order to decide whether this belongs to a high detail or to a low detail area. We denote the variance around a pixel with co-ordinates (x, y) as $\sigma^2_{(x,y),W}$. The subscript W denotes the neighborhood used in order to calculate this variance. If this is a rectangular window of size $(2W + 1)(2W + 1)$ symmetrically placed around the pixel of interest, then

$$\sigma^2_{(x,y),W} = \frac{1}{(2W+1)} \sum_{i=-W}^{i=+W} \sum_{j=-W}^{j=+W} f(x + i, y + j)^2 - m^2_{(x,y),W}$$

Therefore,

If $m_{(x,y),W} > T_1$ or $\sigma^2_{(x,y),W} > T_2$ then use a small low-pass spatial mask for noise removal.

Otherwise, if $m_{(x,y),W} < T_1$ and $\sigma^2_{(x,y),W} < T_2$ then use a large low-pass spatial mask for noise removal. **[5]**

(c)

$f(x, y)\} = f(x, y) * \delta(x, y)$ and $mean\{f(x, y)\} = f(x, y) * w(x, y)$. Therefore,

$$g(x,y) = \begin{cases} 1, & f(x,y) * (\delta(x,y) - w(x,y)) > T \\ 0, & otherwise \end{cases}$$

0	0	0	0	0
0	0	0	0	0
0	0	1	0	0
0	0	0	0	0
0	0	0	0	0

Figure: $\delta(x, y)$

1/25	1/25	1/25	1/25	1/25
1/25	1/25	1/25	1/25	1/25
1/25	1/25	1/25	1/25	1/25
1/25	1/25	1/25	1/25	1/25
1/25	1/25	1/25	1/25	1/25

Figure: $w(x, y)$

-1/25	-1/25	-1/25	-1/25	-1/25
-1/25	-1/25	-1/25	-1/25	-1/25
-1/25	-1/25	24/25	-1/25	-1/25
-1/25	-1/25	-1/25	-1/25	-1/25
-1/25	-1/25	-1/25	-1/25	-1/25

Figure: $\delta(x, y) - w(x, y)$

[5]

3. (a) (i) Wiener filter in space

$$W = R_{fy}{R_{yy}}^{-1} = R_{ff}H^T(HR_{ff}H^T + R_{nn})^{-1}$$

Estimate for the original image is

$$\hat{f} = R_{ff}H^T(HR_{ff}H^T + R_{nn})^{-1}y$$

Note that knowledge of $R_{ff} = E\{ff^T\}$ and $R_{nn} = E\{nn^T\}$ is assumed.

In frequency domain

$$W(u,v) = \frac{S_{ff}(u,v)H^*(u,v)}{S_{ff}(u,v)|H(u,v)|^2 + S_{nn}(u,v)}$$

$$\hat{F}(u,v) = \frac{S_{ff}(u,v)H^*(u,v)}{S_{ff}(u,v)|H(u,v)|^2 + S_{nn}(u,v)}Y(u,v)$$

The noise variance has to be known, otherwise it is estimated from a flat region of the observed image.

In practical cases where a single copy of the degraded image is available, it is quite common to use $S_{yy}(u,v)$ as an estimate of $S_{ff}(u,v)$. This is very often a poor estimate. **[5]**

(ii) If $S_{nn}(u,v) = 0 \Rightarrow W(u,v) = \frac{1}{H(u,v)}$ which is the inverse filter. If $S_{nn}(u,v) \to 0$

$$\lim_{S_{nn}\to 0} W(u,v) = \begin{cases} \dfrac{1}{H(u,v)} & H(u,v) \neq 0 \\ 0 & H(u,v) = 0 \end{cases}$$

which is the pseudo-inverse filter. **[5]**

(b) (i) The noise is white with variance σ_n^2 and therefore the autocorrelation function of the noise is

$$R_{nn}(k,l) = \underset{(x,y)}{E}\{n(x,y)n(x+k,y+l\} = \sigma_n^2\delta(k,l)$$

and the power spectrum of the noise is σ_n^2.

$$S_{ff}(u,v) = \sum_{k=-\infty}^{\infty}\sum_{l=-\infty}^{\infty} R_{ff}(k,l)e^{-j(uk+vl)} = \sum_{k=-\infty}^{\infty}\sum_{l=-\infty}^{\infty} \rho^{|k|+|l|}\, e^{-j(uk+vl)}$$

$$= \sum_{k=-\infty}^{\infty} \rho^{|k|}\, e^{-juk} \sum_{l=-\infty}^{\infty} \rho^{|l|}\, e^{-jvl}$$

We find that

$$\sum_{n=-\infty}^{\infty} r^{|n|}e^{-j\omega n} = \sum_{n=-\infty}^{-1} r^{-n}e^{-j\omega n} + \sum_{n=0}^{\infty} r^n e^{-j\omega n} = \sum_{n=1}^{\infty} r^n e^{j\omega n} + \sum_{n=0}^{\infty} r^n e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} r^n e^{j\omega n} + \sum_{n=0}^{\infty} r^n e^{-j\omega n} - 1$$

$$= \sum_{n=0}^{\infty} (re^{j\omega})^n + \sum_{n=0}^{\infty} (re^{-j\omega})^n - 1 = \frac{1}{1-re^{j\omega}} + \frac{1}{1-re^{-j\omega}} - 1$$

$$= \frac{2-2r\cos\omega}{1+r^2-2r\cos\omega} - 1 = \frac{1-r^2}{1+r^2-2r\cos\omega}$$

Therefore,

$$S_{ff}(u,v) = \frac{(1-\rho^2)^2}{(1+\rho^2-2\rho\cos u)(1+\rho^2-2\rho\cos v)}$$

and

$$W(u,v) = \frac{(1-\rho^2)^2}{(1-\rho^2)^2+(1+\rho^2-2\rho\cos u)(1+\rho^2-2\rho\cos v)\sigma_n^2}$$ **[5]**

(ii) Develop a method of estimating ρ_1 and ρ_2 from $g(x, y)$.

$$R_{gg}(k, l) = R_{ff}(k, l) + R_{nn}(k, l) = \rho^{|k|+|l|} + \sigma_n^2 \delta(k, l)$$

If we choose two pairs of values for the parameters k, l as $(k, l) = (1,0)$ and $(k, l) = (0,0)$ we have the following relationships:

For $(k, l) = (1,0)$: $R_{gg}(1,0) = \rho$ where $R_{gg}(1,0)$ is estimated from the available data.

For $(k, l) = (0,0)$: $R_{gg}(0,0) = 1 + \sigma_n^2$ where $R_{gg}(0,0)$ is estimated from the available data

[5]

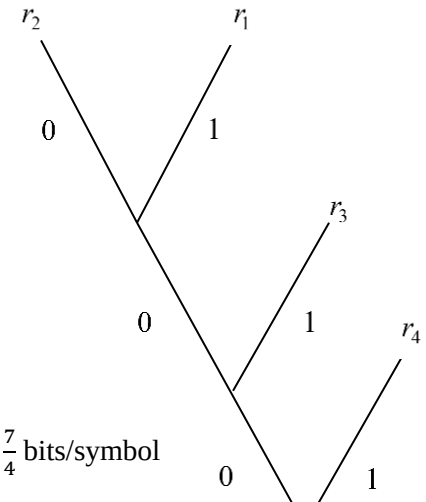
4. (a) (i) The four reconstruction levels are:
$r_1 = 1/8$ with probability $1/8$
$r_2 = 3/8$ with probability $1/8$
$r_3 = 5/8$ with probability $1/4$

$r_4 = 7/8$ with probability ½ **[4]**

(ii) Uniform quantization does not exploit the pdf of the alphabet to be quantized. **[4]**

(iii) The Huffman code is found below. **[4]**

Step 1		Step 2		Step 3	
r_4	1/2	r_4	1/2	r_4	1/2
r_3	1/4	r_3	1/4	$\{r_3,\{r_2,r_1\}\}$	1/2
r_2	1/8	$\{r_2,r_1\}$	1/4		
r_1	1/8				



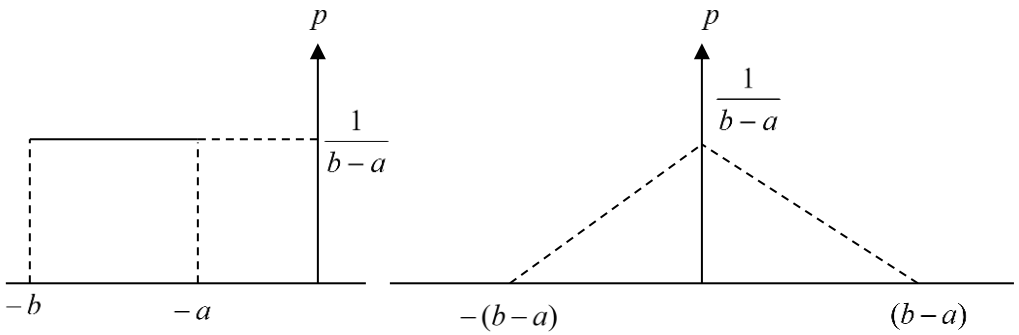
Symbol	*Codeword*
r_1	001
r_2	000
r_3	01
r_4	1

Average number of bits to represent f

$$l_{avg} = \frac{1}{2} + 2\frac{1}{4} + 6\frac{1}{8} = \frac{7}{4} \text{ bits/symbol}$$

(iv) Entropy $H(s) = -\sum_{i=1}^{3} p_i \log_2(p_i) = -(\frac{1}{2}log_2\frac{1}{2} + \frac{1}{4}log_2\frac{1}{4} + \frac{1}{8}log_2\frac{1}{8} + \frac{1}{8}log_2\frac{1}{8})$
$= \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{3}{8} = \frac{4}{8} + \frac{4}{8} + \frac{3}{8} + \frac{3}{8} = \frac{14}{8} = \frac{7}{4}$ bits/symbol
Redundancy = 0bits/symbol
Coding efficiency $H(s)/l_{avg} = 100\%$ **[4]**

(b)



The pdf of $-f(x-1, y-1)$ is shown on the left and the pdf of $g(x, y)$ is shown on the right (convolution of the 2 pdf's). The pdf of $g(x, y)$ is more skewed and therefore more appropriate for symbol encoding. **[4]**